

Data Mining

Lab - 4

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Step 1. Import the necessary libraries

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

print(pd.__version__)
print(np.__version__)
```

2.2.2
1.26.4

Step 2. Import the dataset from this [address](#).

Step 3. Assign it to a variable called chipo.

```
In [2]: chipo = pd.read_csv("https://raw.githubusercontent.com/justmarkham/DAT8/master/data/chipotle.tsv", sep='\t')
chipo
```

Out[2]:

	order_id	quantity	item_name	choice_description	item_price	
	0	1	1	Chips and Fresh Tomato Salsa	NaN	\$2.39
	1	1	1	Izze	[Clementine]	\$3.39
	2	1	1	Nantucket Nectar	[Apple]	\$3.39
	3	1	1	Chips and Tomatillo-Green Chili Salsa	NaN	\$2.39
	4	2	2	Chicken Bowl	[Tomatillo-Red Chili Salsa (Hot), [Black Beans...	\$16.98
	...	...	...	...	...	...
	4617	1833	1	Steak Burrito	[Fresh Tomato Salsa, [Rice, Black Beans, Sour ...	\$11.75
	4618	1833	1	Steak Burrito	[Fresh Tomato Salsa, [Rice, Sour Cream, Cheese...	\$11.75
	4619	1834	1	Chicken Salad Bowl	[Fresh Tomato Salsa, [Fajita Vegetables, Pinto...	\$11.25
	4620	1834	1	Chicken Salad Bowl	[Fresh Tomato Salsa, [Fajita Vegetables, Lettu...	\$8.75
	4621	1834	1	Chicken Salad Bowl	[Fresh Tomato Salsa, [Fajita Vegetables, Pinto...	\$8.75

4622 rows × 5 columns

Step 4. See the first 10 entries

In [3]:

```
chipo.head(10)
```

Out[3]:

	order_id	quantity	item_name	choice_description	item_price	
	0	1	1	Chips and Fresh Tomato Salsa	NaN	\$2.39
	1	1	1	Izze	[Clementine]	\$3.39
	2	1	1	Nantucket Nectar	[Apple]	\$3.39
	3	1	1	Chips and Tomatillo-Green Chili Salsa	NaN	\$2.39
	4	2	2	Chicken Bowl	[Tomatillo-Red Chili Salsa (Hot), [Black Beans...	\$16.98
	5	3	1	Chicken Bowl	[Fresh Tomato Salsa (Mild), [Rice, Cheese, Sou...	\$10.98
	6	3	1	Side of Chips	NaN	\$1.69
	7	4	1	Steak Burrito	[Tomatillo Red Chili Salsa, [Fajita Vegetables...	\$11.75
	8	4	1	Steak Soft Tacos	[Tomatillo Green Chili Salsa, [Pinto Beans, Ch...	\$9.25
	9	5	1	Steak Burrito	[Fresh Tomato Salsa, [Rice, Black Beans, Pinto...	\$9.25

Step 5. What is the number of observations in the dataset?

In [4]:

```
# Solution 1

chipo.shape[0]
```

Out[4]: 4622

In [5]:

```
# Solution 2

chipo.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4622 entries, 0 to 4621
Data columns (total 5 columns):
#   Column              Non-Null Count  Dtype
---  -
0   order_id            4622 non-null   int64
1   quantity            4622 non-null   int64
2   item_name           4622 non-null   object
3   choice_description  3376 non-null   object
4   item_price          4622 non-null   object
dtypes: int64(2), object(3)
memory usage: 180.7+ KB
```

Step 6. What is the number of columns in the dataset?

```
In [6]: chipo.shape[1]

Out[6]: 5
```

Step 7. Print the name of all the columns.

```
In [7]: chipo.columns

Out[7]: Index(['order_id', 'quantity', 'item_name', 'choice_description',
              'item_price'],
              dtype='object')
```

Step 8. How is the dataset indexed?

```
In [8]: chipo.index

Out[8]: RangeIndex(start=0, stop=4622, step=1)
```

Step 9. Number of Unique Items ?

```
In [9]: chipo['item_name'].nunique()

Out[9]: 50
```

Step 10. Which was the most-ordered item?

```
In [10]: chipo.groupby('item_name').sum().sort_values(['quantity'],ascending=False).head(1)

Out[10]:
```

	order_id	quantity	choice_description	item_price
	item_name			
	Chicken Bowl	713926	761 [Tomatillo-Red Chili Salsa (Hot), [Black Beans...	16.9810.98 11.258.75 8.4911.25 \$8.75 ...

Step 11. How many items were orderd in total?

```
In [11]: chipo['quantity'].sum()

Out[11]: 4972
```

Step 12. Turn the item price into a float

Step 12.a. Check the item price type

```
In [12]: chipo.dtypes['item_price']
```

```
Out[12]: dtype('O')
```

Step 12.b. Create a lambda function and change the type of item price

```
In [13]: dollarizer= lambda x:float(x[1:-1])
chipo.item_price=chipo.item_price.apply(dollarizer)
```

Step 12.c. Check the item price type

```
In [14]: chipo.dtypes['item_price']
chipo.item_price
```

```
Out[14]: 0      2.39
1      3.39
2      3.39
3      2.39
4     16.98
...
4617   11.75
4618   11.75
4619   11.25
4620    8.75
4621    8.75
Name: item_price, Length: 4622, dtype: float64
```

Step 14. How much was the revenue for the period in the dataset?

```
In [15]: print('Revenue was: $'+str((chipo.item_price*chipo.quantity).sum()))
```

```
Revenue was: $39237.02
```

Step 15. How many orders were made ?

```
In [16]: chipo.order_id.nunique()
```

```
Out[16]: 1834
```

Step 17. How many different choice descriptions are there?

```
In [17]: chipo.choice_description.nunique()
```

```
Out[17]: 1043
```

Step 18. What items have been ordered more than 100 times?

```
In [18]: chipo.groupby('item_name').sum()['quantity'].loc[lambda x: x>100]
```

```
Out[18]: item_name
Bottled Water      211
Canned Soda        126
Canned Soft Drink  351
Chicken Bowl       761
Chicken Burrito     591
Chicken Salad Bowl 123
Chicken Soft Tacos  120
Chips               230
Chips and Fresh Tomato Salsa 130
Chips and Guacamole 506
Side of Chips       110
Steak Bowl          221
Steak Burrito       386
Name: quantity, dtype: int64
```

Step 19. What is the average revenue amount per order?

```
In [19]: # Solution 1

(chipo.item_price*chipo.quantity).sum()/chipo.order_id.nunique()
```

Out[19]: 21.39423118865867

```
In [20]: # Solution 2

chipo['revenue']=chipo.item_price*chipo.quantity
chipo.groupby('order_id').revenue.sum().mean()
```

Out[20]: 21.39423118865867